

PEM 2023 Year 11 Mathematics Standard Yearly Examination

SUGGESTED SOLUTIONS

Section 1 Multiple Choice Answer Key

Question	Answer	Outcomes Assessed	Targeted Performance Bands
1	<i>D</i>	<i>MS11–1</i>	3–4
2	<i>B</i>	<i>MS11–2</i>	2–3
3	<i>C</i>	<i>MS11–1</i>	3–4
4	<i>C</i>	<i>MS11–5</i>	3–4
5	<i>A</i>	<i>MS11–1</i>	4–5
6	<i>C</i>	<i>MS11–6</i>	4–6
7	<i>A</i>	<i>MS11–9</i>	2–3
8	<i>D</i>	<i>MS11–6</i>	4–5
9	<i>B</i>	<i>MS11–4</i>	3–4
10	<i>A</i>	<i>MS11–3</i>	4–5
11	<i>A</i>	<i>MS11–9</i>	4–5
12	<i>B</i>	<i>MS11–8</i>	3–5

Section II (Total 68 marks)**Question 13 A1**

Outcomes Assessed: MS11– 1

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct x - value	2
Makes progress towards the solution	1

Sample Answer:

$$5 - 5x = 3x - 3$$

$$\therefore 8 = 8x$$

$$\therefore x = 1$$

Question 14 (a) M1.2

Outcomes Assessed: MS11– 4

Targeted Performance Band: 3-4

Criteria	Marks
Uses correct formula for volume	1
Provides correct volume in cubic meters	2
Provides correct solution in L	3

Sample Answer:

$$V = \frac{1}{2} \times 24 \times 7 \times 22 = 1848 \text{ m}^3$$

$$1 \text{ m}^3 = 1000 \text{ L so } V = 1848000 \text{ L}$$

Question 14 (b) M1.2

Outcomes Assessed: MS11– 4

Targeted Performance Band: 4-5

Criteria	Marks
Makes progress towards answer by finding the area to paint	1
Provides correct solution	2

Sample Answer:

2 sides

$$2 \times \left(\frac{1}{2} \times 24 \times 7 \right) = 168;$$

Base

$$22 \times \text{hypotenuse}(c) = 22 \times 25 = 550$$

$$c^2 = 7^2 + 24^2, c = 25$$

$$\text{Total area } 168 + 550 = 718 \text{ m}^2$$

$$\text{Cost } 12 \times 718 = \$8616$$

Question 15 (a) F1

Outcomes Assessed: MS11– 5

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	1

Sample Answer:

$$43700 - 37000 = 6700$$

$$\frac{6700}{43700} \times \frac{100}{1} = 15.33\%$$

Question 15(b)

Outcomes Assessed: MS11– 5

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	1

Sample Answer:

3 years decrease is 6700

$$\frac{6700}{3} = \$2233.33 \text{ per year}$$

Question 15(c)

Outcomes Assessed: MS11– 5

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	1

Sample Answer:

$$60\% \times 1690 = \$1014$$

Question 16 M1.1

Outcomes Assessed: MS11– 3

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct answer	1

Sample Answer:

$$52.79 \times (2.3 \times 10^3) = 121417 \approx 120000 \text{ to 2 significant figures}$$

Question 17 A1

Outcomes Assessed: MS11– 6

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct substitution into formula	1
Provides correct solution	2

Sample Answer:

$$D = \frac{18 \times 400}{140} = 51.43 \approx 51 \text{ mg}$$

Question 18 (a) A2

Outcomes Assessed: MS11– 1

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct gradient	1

*Sample Answer:*Gradient is $-\frac{2}{3}$ **Question 18 (b) A2**

Outcomes Assessed: MS11– 1

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct x - value	1

Sample Answer:

$$y = 0, \quad 2x - 1 = 0$$

$$2x = 1$$

$$x = \frac{1}{2}$$

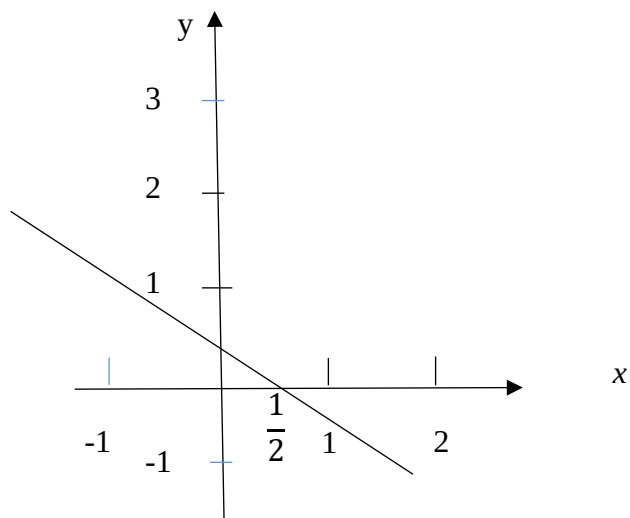
Question 18 (c) A2

Outcomes Assessed: *MS11– 3*

Targeted Performance Band: 3-4

Criteria	Marks
Draws the correct clearly labelled graph	2
Shows an intercept or gradient	1

Sample Answer:



Question 19 (a) M1.2

Outcomes Assessed: MS11– 4

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	2
Makes progress towards the solution by use of the surface area formula	1

Sample Answer:

$$\begin{aligned}
 SA &= \pi r(r + \sqrt{h^2 + r^2}) \\
 &= 9\pi(9 + \sqrt{121 + 81}) \\
 &\approx 656.32\text{cm}^2
 \end{aligned}$$

Question 19 (b) M1.2

Outcomes Assessed: MS11– 4

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct solution	1

Sample Answer:

$$120 \div 6 = 20 \text{ sheets of cardboard}$$

Question 19 (c) M1.2

Outcomes Assessed: MS11– 4

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	1

Sample Answer:

$$6 \times 656.32 = 3937.92$$

$$6300 - 3937.92 = 2362.08\text{cm}^2$$

Question 20 (a) F1.1

Outcomes Assessed: MS11– 2

Targeted Performance Band: 3-4

Criteria	Marks
Provides 2 correct facts	2
Provides 1 correct fact	1

Sample Answer:

More data on right side therefore positively skewed.

Median relatively centred.

Large spread.

Question 20 (b) A1

Outcomes Assessed: MS11– 6

Targeted Performance Band: 3-4

Criteria	Marks
Provides both range and IQR	2
Finds either the range or IQR	1

Sample Answer:

Range $90 - 15 = 75$

IQR $66 - 25 = 41$ or $67 - 25 = 42$ acceptable

Question 20 (c) A1

Outcomes Assessed: MS11–6

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	1

Sample Answer:

43 or 44 acceptable

Question 21(a) S2

Outcomes Assessed: MS11-8

Targeted Performance Band: 2-3

Criteria	Marks
Progress towards correct answer	1
Provides correct probability	2

Sample Answer: $P(\text{yellow}) = 1 - 0.81 = 0.19$ **Question 21(b)**

Outcomes Assessed: MS11-8

Targeted Performance Band: 3-4

Criteria	Marks
Finds probability of a purple or red lolly	1
Provides correct expected value	2

Sample Answer:

$$p = 0.15 + 0.2 = 0.35$$

$$E = np = 35 \times 0.35 = 12.25$$

Therefore, he would expect about 12 lollies to be his favourites

Question 22 M1.3

Outcomes Assessed: MS11-3

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	3
Makes progress towards answer	2
Provides one correct calculation	1

Sample Answer:

Energy used to run the pool per week is $2000 \times 2.5 \times 6 = 30000Wh$

$30000Wh = 30kWh$ per week

Cost per year is $30 \times 52 \times 12.199 \text{ cents} = 19030.44 \text{ cents} = \190.30

Question 23 (a) M1.2

Outcomes Assessed: MS11-4

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	2
Shows use of trapezoidal rule	1

Sample Answer:

$$\text{Total Area} = \frac{18}{2}(23+15) + \frac{18}{2}(15+19) = 648\text{m}^2$$

Question 23 (b)

Outcomes Assessed: MS11-4

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct solution in m ³	2
Shows method to calculate volume	1

Sample Answer:

$$\text{Volume} = 648 \times 200 \div 10 \div 100 = 129.6\text{m}^3$$

Question 24 (a) S1.1

Outcomes Assessed: MS11-10

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct description	1

Sample Answer:

36

Question 24 (b)

Outcomes Assessed: MS11-7

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution	1

Sample Answer:

$36/2 = 18$ so 18th and 19th score are in the middle which are 121 and 122 respectively.

So median is 121.5

Question 24 (c)

Outcomes Assessed: MS11-7

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct solution	1

Sample Answer:

$$\frac{22}{36} \times \frac{100}{1} = 61.1\%$$

Question 25 (a) M2

Outcomes Assessed: MS11-3

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution for total time	2
Makes progress towards calculating the correct total time	1

Sample Answer:

10:04 to 11:48 is 1 hr 44 mins.

11:49 to 1:23 is 1hr 34 mins.

Total time of partial eclipse is 3 hr 18 mins

Question 25 (b) M2

Outcomes Assessed: MS11-3

Targeted Performance Band: 4-6

Criteria	Marks
Provides correct solution	2
Gives either start or finish time in Sydney	1

Sample Answer:

If WA is 2 hours ahead of Sydney

11:48am is 1:48pm start of total eclipse in Sydney time.

11:49am is 1:49pm end of total eclipse in Sydney time.

Question 26 (a) F1.2

Outcomes Assessed: MS11-5

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct solution of refund and how much	4
Calculates tax payable correctly	3
Uses correct bracket for Taxable income	2
Calculate Taxable Income correctly	1

Sample Answer:

Taxable income is $\$68500 - 750 - 200 = 67550$

Tax bracket is $\$5092 + 32.5$ cents for each dollar over $\$45000$

Tax = $5092 + 32.5\% \times (67550 - 45000)$

= $\$12420.75$

Medicare levy = $2\% \times 67550 = \$1351$

Total tax = $1351 + 12420.75 = \$13771.75$

Refund of $13771.75 - 12420.75 = \$1628.25$

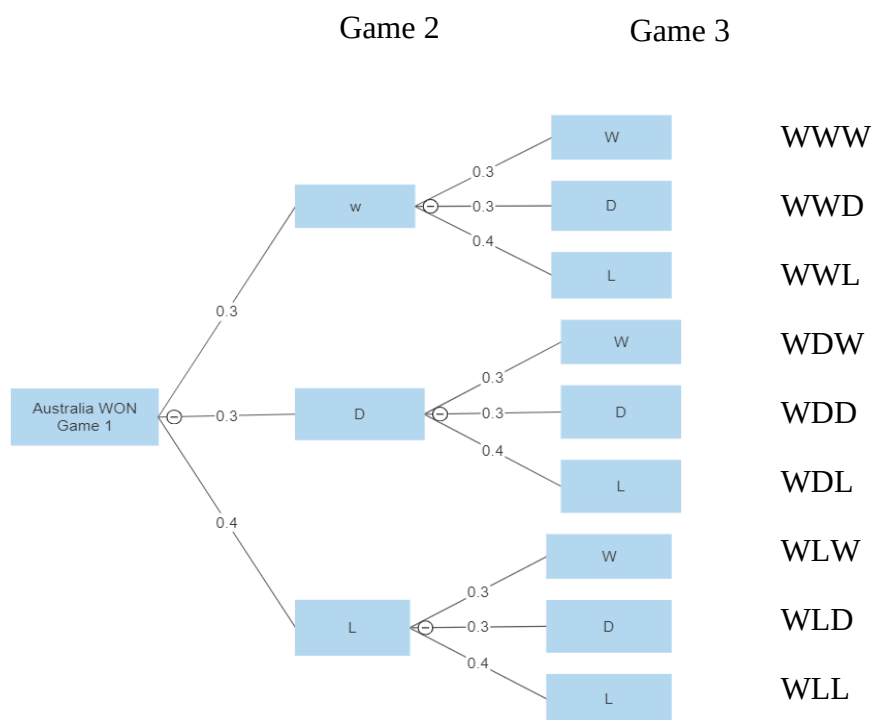
Question 27 (a) MS1.2

Outcomes Assessed: MS11-8

Targeted Performance Band: 3-4

Criteria	Marks
Correct tree diagram	1
Provides correct tree diagram with list of possible outcomes	2

Sample Answer:



Question 27 (b) MS1.2

Outcomes Assessed: MS11-8

Targeted Performance Band: 3-4

Criteria	Marks
Makes progress towards answer	1
Provides correct probability	2

Sample Answer: 51% or 0.51

WWW or WWD or WWL or WDW or WLW = 0.09 + 0.09 + 0.12 + 0.09 + 0.12 = 0.51

Question 28 (a) S1.2

Outcomes Assessed: MS11-7

Targeted Performance Band: 4-5

Criteria	Marks
Provides correct number of students	1

Sample Answer: $8 + 21 + 17 + 5 + 2 = 53$ **Question 28 (b) S1.2**

Outcomes Assessed: MS11-11

Targeted Performance Band: 3-5

Criteria	Marks
Provides correct solution	1

*Sample Answer:*Most frequent score falls into the class group $10 < t \leq 20$ **Question 28 (c) S1.2**

Outcomes Assessed: MS11-10

Targeted Performance Band: 3-5

Criteria	Marks
Provides correct solution	2
Makes some progress towards the solution	1

Sample Answer:

$$17 + 5 + 2 = 24, \frac{24}{53} \times \frac{100}{1} = 45.28\%$$

Question 28 (d) S1.1

Outcomes Assessed: MS11-2

Targeted Performance Band:

Criteria	Marks
Provides correct solution	1

Sample Answer:

$$\frac{53+1}{2} = 27^{\text{th}} \text{ score}$$

which falls into the group

$$10 < t \leq 20$$

Question 29 (a) S1.1

Outcomes Assessed: MS11-10

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	1

Sample Answer:

79%

Question 29 (b) S1.1

Outcomes Assessed: MS11-10

Targeted Performance Band: 3-4

Criteria	Marks
Provides one of the most common complaints	1
Provides both most common complaints	2

Sample Answer:

Installation and Software faults

Question 30 (a) F1.1

Outcomes Assessed: MS11-5

Targeted Performance Band: 2-3

Criteria	Marks
Provides correct solution	1

Sample Answer:

$$3\% \times \$50000 = \$1500$$

Question 30 (b) F1.1

Outcomes Assessed: MS11-5

Targeted Performance Band: 3-4

Criteria	Marks
Makes some progress towards the solution	1
Correctly calculates values in table only	2
Provides correct solution	3

Sample Answer:

Age	Amount in account
67	50000
68	51500
69	$51500 + 1500 = 53000$
70	$53000 + 1500 = 54500$
71	$54500 + 1500 = 56000$
72	$56000 + 1500 = 57500$
73	$57500 + 1500 = 59000$

Fred will have \$59000 in his account at age 73.

Question 30 (c) F1.3

Outcomes Assessed: MS11-5, MS11-10

Targeted Performance Band: 3-4

Criteria	Marks
Provides correct solution	2
Makes some progress towards the solution	1

Sample Answer:

$$I = Prn$$

$$\begin{aligned} I &= 12000 \times 7\% \times 10 \\ &= \$8400 \end{aligned}$$